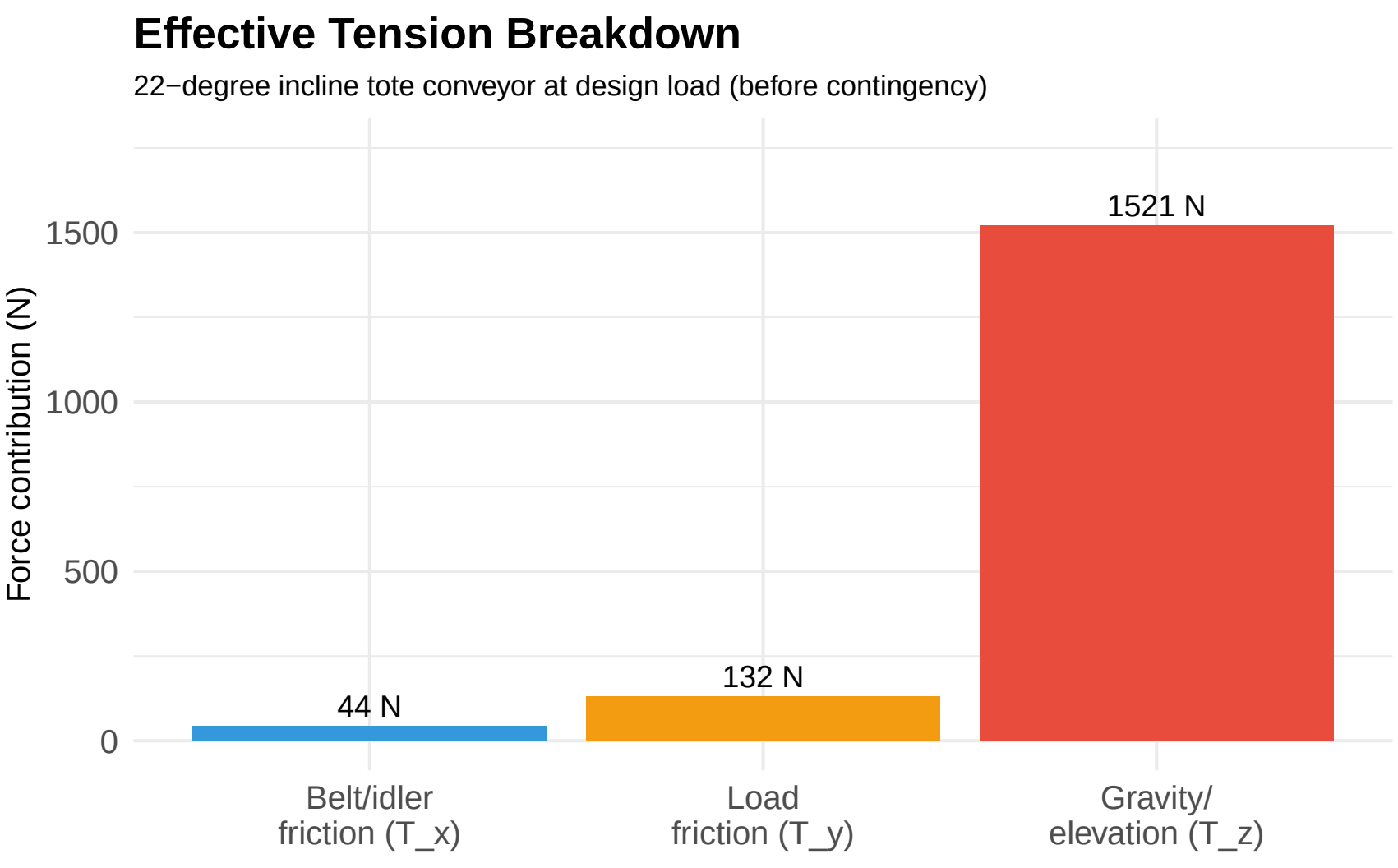
Effective Tension Breakdown
22−degree incline tote conveyor at design load (before contingency)
Force contribution (N)
1500
1000
500
0
44 N
132 N
1521 N
Belt/idler friction (T_x)
Load friction (T_y)
Gravity/ elevation (T_z)